

Gao Bin

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EDUCATION

Sep 2021 — Jun 2025 **Shanxi University of Finance and Economics** *Bachelor in Applied Statistics*

- **GPA:** 3.81/4, 91.4/100
- **Honors:** National Scholarship ,Merit Student several times and Outstanding Graduates
- **Relevant Coursework:** Mathematical Statistics/Introduction to Econometrics (Bilingual)/ Game Theory (Bilingual) /Deep Learning

Sep 2025 — **Minzu University of China(985)** *Master in Applied Statistics*

- **GPA:** 92.2/100
- **Honors:** First-Class Academic Scholarship
- **Relevant Coursework:** Machine Learning Methods and Applications(98) /Modern optimization methods(92)

PUBLICATIONS

Peer review

[1] Bin Gao, Zihao ZHENG, Chuanhua Wei. Flow-GeoShapley: A Geographical Interpretable Machine Learning Method for Spatial Flow Data. *Geographical Systems*. 2026.

[2] Bin Gao, Xinyu Nie, Jinhua Zhang, Chuanhua Wei. Based on interpretable machine learning quantifying nonlinear contributions and spatial effects of urban shrinkage on reducing pollution and carbon emissions. *Environmental Development*. 2026.

Work paper

[1] Bin Gao, Chuanhua Wei. Simultaneous estimation of model parameters in flow data.2026+

[2]Bin Gao, Chuanhua Wei. A novel approach to quantify non- linear and mediate effect in psychological intensive longitudinal data. 2026+.

[3] Bin Gao, Xinxin Dan, Lanlan Xiang, Chuanhua Wei. What Attracts Highly Educated Talents to Settle Down? — A Multi-Model Fusion Analysis Based on Explainable Machine Learning.2026

Research Interests: spatial flow; model interpretation in space;

AWARED & FELLOWSHIPS

2026	China graduate Mathematical Contest in Modelling Third prize (7%)
2025	Energy Economics Academic Ideas Competition Second prize (5%)
2024	National Applied Statistics Graduate Degree Case Competition First Prize in North China Division
2024	Energy Economics Academic Ideas Competition Third prize (7%)
2024	National Undergraduate Market Research and Analysis Competition Third prize (10%)
2024	China Undergraduate statistical Contest in Modelling Third prize (5%)
2024	"Challenge Cup" National Contest Provincial Gold Award (15%)
2023	National Undergraduate Market Research and Analysis Competition Provincial First Prize
2023	China Undergraduate statistical Contest in Modelling Third prize (8%)
2023	China Undergraduate Mathematical Contest in Modelling Provincial First Prize (20%)
2022-2024	National Encouragement Scholarship for Top 10% students with academic performance(twice)
2022-2024	The First Prize Scholarship for the student with the first comprehensive quality (five times)

ACADEMIC EXPERIENCE

2026 — **Flow-GeoSHAP: An Explainability Method for Extracting Spatial Effects from Machine Learning Models of Flow Data**

Project leader

- Propose and optimize the FLOW-GeoShapley method framework. detail package information in <https://github.com/GaoBin88/FLOW-GeoSHAP-supplyment>.

2023 — **Spatiotemporal Evolution and Driving Force Analysis of Inclusive Green Growth in Shanxi Province: Based on the Perspective of Chinese Modernization** (Key Project of Statistical Science Research in Shanxi Province, NO.2023LZ035)

first participant

- Built a three-dimensional measurement system for social well-being, green and sustainable development and economic growth connecting with a new theory of Chinese-style modernization

- Collected and cleaned the data needed for 2007-2022 to ensure the timeliness and validity of the study using R studio and conducted Spatial Q statistics method and Moran Index using Geoda to measure spatial heterogeneity and association and demonstrated on map
- Finished work paper as the first author which wonned the third price in 9th China undergraduate statistical contest in modelling

Advisor: prof. Liqin Zhao, Shanxi Univ Finance & Econ, Sch Stat

2023 — **Accounting of informal employment in Shanxi Province under the digital economy** (Key Project of Statistical Science Research in Shanxi Province, NO.2023LZ006)
first participant

2022 — **Accounting of informal employment under the digital economy in the view of family**(Key Project Funded by national social science foundation of China, NO. 21BTJ012)
Key participant

- Conducted literature retrieval and data collection from database of China time use survey, and illustrated initial tendencies at Origin about duration of employment, etc.
- Sorted out informal employment accounting models based on energy development tendencies in Shanxi Province

Advisor: prof. Yanping Gao, Shanxi Univ Finance & Econ, Sch Stat

2022-2023 **Research of food intangible cultural heritage based on blockchain: an example of flowery pasta in Huan tai county** (National Undergraduate Innovation Training Program, No.202210125028)
Project leader

- Designed of questionnaires and questionnaire quality monitoring methods, such as using 3σ principles to exclude invalid questionnaires and organized of field research and questionnaire distribution which returned 253 questionnaires, of which 193 were valid
- Designed UI interface for food-based intangible cultural heritage sales based on blockchain theory using PIXSO

Advisor: prof. Zhu Bo, Shanxi Univ Finance & Econ, Sch Stat

RELEVANT RESEARCH SKILLS

- **Administrative experience:** Hosted and participated in many team competitions and projects, good at encouraging team members and active atmosphere
 - **Technology skills:** Python (Expert), SPSS(Expert), Stata (Expert), R (Intermediate), ArcGIS(excellent), GeoDa(excellent); Latex. Learning ability from top journal literature to optimize model and produce data presentation figures in R and Origin, such as bcf package, a new model raised in 2020
 - **Languages:** Mandarin Chinese (native); English (conversational)
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PROFESSIONAL & LEADERSHIP EXPERIENCE

Bureau of Statistics
Census takers
Winter 2023

Teamwork, communication skill and Patient data processing style

- Collected year-end asset data for institutions as a connection
- Calibrated data between field survey and ledger about enterprises above designated size

Data marketing, cleaning and processing capabilities

Red Digital China
Data analyst
Apr - Oct 2022

- Operated and analyzed Kelly Hoppen's social medias data marketing to portray the portrait of the target group
- Extracted website keywords and categorize them into four channels such as long-tail and consumer conversion to assist in the next monthly marketing direction

Department of Justice
financial management
Winter 2022

Data retrieval and programming capabilities

- Skilled in using PyCharm, anaconda and other platforms to analyze the month's website visit traffic and Baidu SEM and other marketing situations

- Invoked Baidu's API interface to automatically generate monthly reports of GGB company

Excellent financial management performance

- Responsible for handing over the capital data with the bank and completing the year-end salary payment